THE UNITED REPUBLIC OF TANZANIA
ORGANIZATION OF HEADS OF NON-GOVERNMENT SECONDARY SCHOOLS-TANZANIA (OHONGSS-T)
MWANZA & KIGOMA REGIONS
FORM FOUR MOCK EXAMINATION-MAY, 2023

041 **BASIC MATHEMATICS**

Time: 3 Hours **Tuesday, 23th May, 2023 A.M**

Instructions

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1. This paper consists of section **A** and **B** with a total of **fourteen (14)** questions.
2. Answer **all** questions in both sections.
3. Each question in section **A** carries **six (6)** marks while each question in section **B** carries **ten (10)** marks.
4. All necessary working and answers for each question done must be shown clearly.
5. Mathematical tables and non – programmable calculators may be used.
6. All communication devices and any unauthorized materials are not allowed in the examination room.
7. Write your Examination **Number on every page** of your answer booklet (s).
8. Where necessary, use the following constants:
   - Pie, $\pi = 3.14$
   - Radius of the Earth, $R_e = 6370km$

This paper consists of **7** printed pages

SECTION A (60Marks)

Answer **ALL** questions in this section.

1. (a) Three wooden rods of lengths 168cm, 72cm and 90 cm are cut into pieces of the same lengths. Find the greatest possible length for these pieces of roads if no wooden rod is left without cut.

   (b) The enrolment of the Form one students in a certain secondary school was 500 last year. The enrolment this year at the school is 105% of the last year's enrolment. Find the enrolment of this year at the school.

2. (a) The amount of money accumulated in somebody's fixed account at the CRDB bank is given by: $A = P\left(1 + \frac{R}{100}\right)^n$.

   Where **P** is the principal compounded annually.

   R is the annual interest rate.

   n is the number of years the money is deposited.

   Make **"n"** as the subject of this formula.

   (b) Show that the sum of $\frac{1}{2-\sqrt{3}}$ and $\frac{1}{2+\sqrt{3}}$ is a rational number.

3. (a) A survey of 160 house holders, each one of whom kept a cat or a dog or both, showed that 95 kept cats and 80 kept dogs. How many kept both a cat and a dog?

   (b) A family has three children. By using a tree diagram, find the probability that the family has:

   (i) All girls

   (ii) One girl and two boys.

4. (a) The coordinates of a garden plot with corner points A, B, C and D are (0,0), (3, -2), (2,3) and (12,5) respectively in metres. Find:

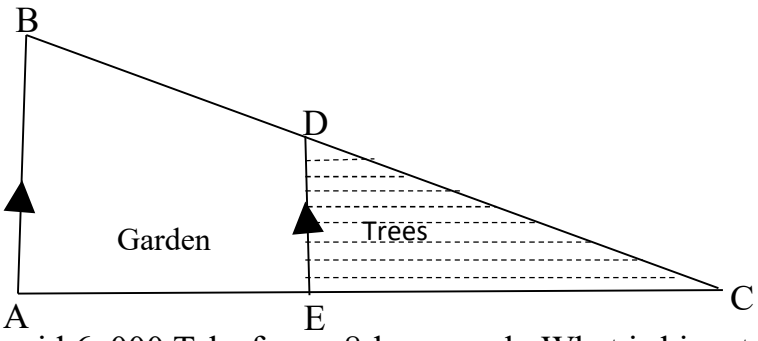
   (i) The slope of the side BD.

   (ii) The coordinates of the mid –point of the diagram BC.

   (b) The position of ship with respect to a fixed origin 0 is given by: $\underset{\sim}{r} = (0,10) + t(0.4, -0.3)$, where scalar **t** is the time in hours and distances are measured in kilometres. Calculate the position of the ship when t = 20.

5. (a) Find the length of a side and the perimeter of a regular nine –sided polygon inscribed in a circle of radius 10cm.

(b) A man owns a triangular farm ABC. He decides to divide it into two portions ABDE and CDE of which are to be reserved for a garden and planting trees respectively. The portion CDE reserved for trees is shaded and has the area of $720m^2$. If CE = 50m, CA= 100m and AB is parallel to ED as shown in the figure below, calculate the area of the portion ABDE reserved for the garden.



6. (a) (i) A labour is paid 6, 000 Tshs for an 8-hour work. What is his rate of pay?

(ii) Plant A grows 15 cm in 8 days, plant B grows 12cm in 5 days, and plant C grows 18cm in 11 days. Which plant is growing the slowest? Give a reason.

(b) The numbers of workers needed to repair a road is inversely proportional to the time taken. If 12 workers can finish the repair in 10 days,

(i) Formulate the equation involving the number of workers, **w**, and the time taken, **t**.

(ii) How long will 30 workers take to finish the repair of the same road?

7. (a) A teacher distributed a number of sweets to the three pupils in her baby class in the ratio 3:5:7. John was a pupil in the class who got the smallest share of only 25 sweets. How many sweets did the teacher have altogether before distributing them to those three pupils in the class?

(b) (i) Explain the function of the trading account.

(ii) The ledger of Mr. Festo had the following information:

Sales	Tsh. 38,000.
Opening stock	Ths. 8,000.
Purchases	Ths. 25,000.
Electricity bill	Tsh. 4,000.
Discount allowed	Ths. 2,000.
Closing stock	Tsh. 5,000.

Determine the Gross profit and Net profit/loss of Mr. Festo.

8. (a) A woman saves 40, 000 Tshs in her first year of starting a job. Each year she is able to save 5,000 Tshs more than in the previous year. Formulate an expression for how much she saves in the nth year of the job.

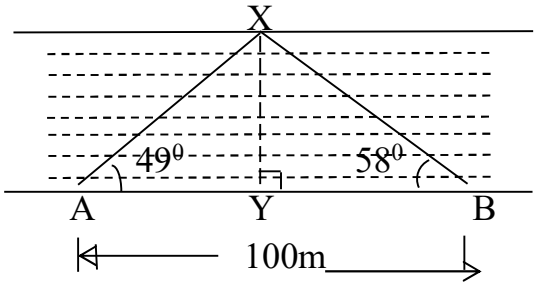
(b) On the first day of a training programme, Hamisi runs 6km. This distance increases by 10% each day. What is the total distance he has run after 12 days?

9. (a) Given that $CosA = \frac{4}{5} and\ A$ is an acute angle. Find the value of:

(i) $\sin A$

(ii) $\frac{TanA-CosA}{1+cosA}$

(b) Mwakami wishes to find the width of $\overline{XY}$ of the Malagalasi river. She measures a distance $\overline{AB}$ = 100m along the bank of the river. She observes that a point X on the other bank of the river is such that $X\hat{A}B = 49^0$ and $X\hat{B}A = 58^0$ as shown in the figure below.



How wide is the river?

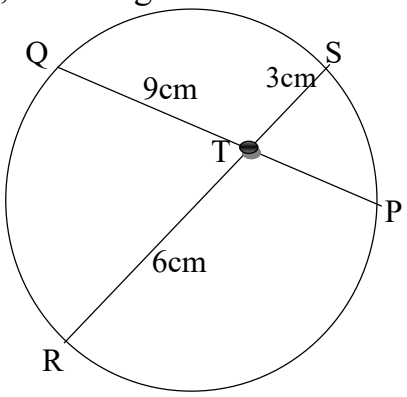
10. (a) A man is 4 times as old as his son. If the product of their ages is 196 years, how old is the son?

(b) The length of a classroom floor is 4m more than the width and the area is $221m^2$. Find the dimensions of the floor.

SECTION B (40Marks)

Answer **all** questions in this section

11. (a) In the figure below, the chords PQ and RS cross at point T. If TR = 6cm, TS = 3cm and TQ =9cm, how long is TP?



(b) The heights of a group of 30 children, measured to the nearest centimetres are as shown in the table below:

Heights(cm)	90-99	100-109	110-119	120-129	130-139	140-149
Number of children	2	5	7	10	5	1

(i) Prepare a frequency distribution including the columns of class boundaries or limits and a cummulative frequency.

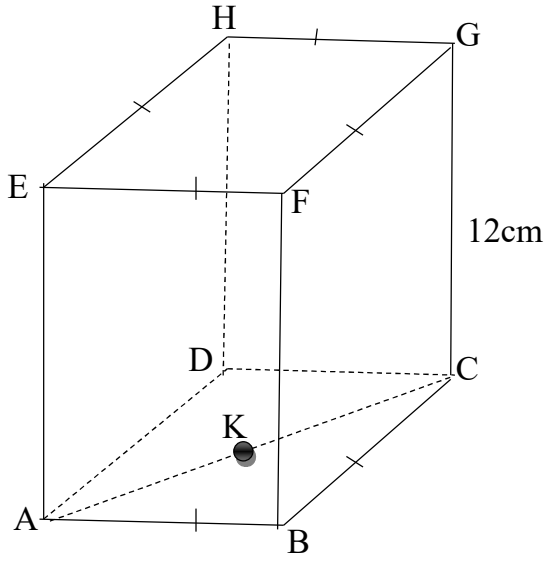
(ii) Draw a cummulative frequency curve and use it to estimate the median height.

12. (a) The figure below is a cuboid with a square base ABCD. If AC = 5cm, CG = 12cm and K is the mid –point of AC, calculate:

(i) The length AG.

(ii) The angle AG makes with the base ABCD.

(iii) The volume of the cuboid.



(b) A plane starts at $(20^0S,\ 30^0E)$ and flies north for a distance 4,000km. Find its new latitude and longitude.

13.(a) In a certain shop, 5 shirts and 2 pairs of shorts cost 27, 000 Tshs. Two shirts and 6 pairs of short cost 29,000 Tshs.

(i) How much is the cost of a shirt and a pair of shorts?

(ii) How much is the total cost of 8 shirts and 5pairs of shorts?

(b) The function f is defined as:

$$f(x) = \begin{cases} -2\ if x < 0 \\ 0\ if\ x = 0 \\ 2\ if\ x\ > 0 \end{cases}$$

Draw the graph of $f(x)$ and from it:

(i) Find the value of $f(7)$

(ii) State the range of $f(x)$

14. (a) Let $A = \{Tanzania, United\ states,\ United\ kingdom, France\}$.
$B = \{dollars,\ shillings,\ pounds, euro\}$.

(i) Draw an arrow diagram to represent the relation between A and B, assuming that both United Kingdom and France also use the currency "euro".

(ii) Is the relation a function or not? Give a reason.

(b) A doctor prescribes that in order to obtain an adequate supply of vitamins A and C, his patient shall have portions of food type 1 and food of type 2. The number of units of Vitamin A and C are given in the following table

	Nutrients	
Food	Vitamin A	Vitamin C
F_1	3	2
F_2	1	7

The doctor prescribes a minimum of 14 units of vitamin A and 21 units of vitamin C. What are the least number of food type 1 and type 2 that will fit the doctor's prescription?